Unit-II
Question & Answers

1. Show that $(A \cap B) \cup C = A \cap (B \cup C)$ iff $C \subseteq A$

Solution :

Given that $(A \cap B) \cup C = A \cap (B \cup C)$

Let $x \in C$
Therefore, $x \in (A \cap B) \cup C$
Implies $x \in A \cap (B \cup C)$ (Since, $(A \cap B) \cup C = A \cap (B \cup C)$)
Therefore, $x \in A$

Therefore, $C \subseteq A$
Therefore , $(A \cap B) \cup C = A \cap (B \cup C)$ implies $C \subseteq A \rightarrow (1)$

Given that $C \subseteq A$

$(A \cap B) \cup C = \{ x \mid x \in (A \cap B)$ or $x \in C \}$

$= \{ x \mid (x \in A$ and $x \in B)$ or $x \in C \}$

$= \{ x \mid (x \in A$ or $x \in C)$ and $(x \in B$ or $x \in C)$

$= \{ x \mid (x \in A$ or $x \in A)$ and $(x \in B \cup C))$ (since, $C \subseteq A$)

$= \{ x \mid (x \in A)$ and $(x \in B \cup C)$

$= A \cap (B \cup C)$

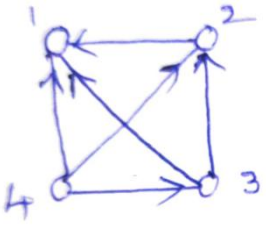
Therefore, $C \subseteq A$ implies $(A \cap B) \cup C = A \cap (B \cup C) \rightarrow (2)$

From (1) and (2) ,

$(A \cap B) \cup C = A \cap (B \cup C)$ iff $C \subseteq A$

2. Let X = { 1,2,3,4} and R = {(x,y) / x > y}. Draw the graph of R and also give its matrix.

Solution : Let X = { 1,2,3,4} and R = {(x,y) / x > y}
R = { (2,1) , (3,1) , (3,2) , (4,1) , (4,2) , (4,3) }
Graph and Matrix :

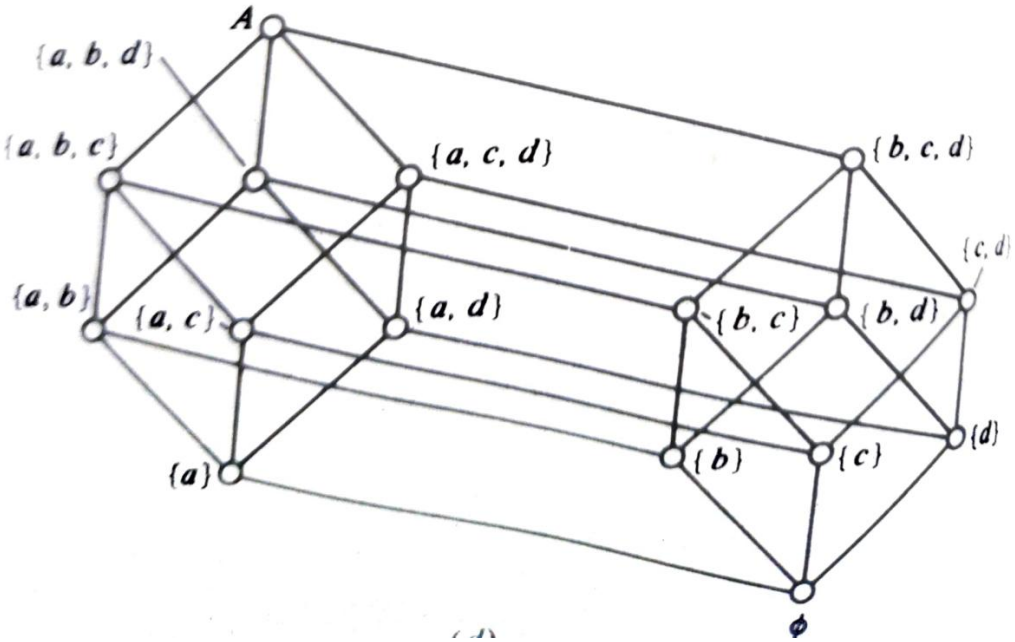


$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{pmatrix} \end{array}$$

3. Let A be a given finite set and P(A) its power set. Draw Hasse diagram of $(P(A), \subseteq)$ for $A = \{a,b,c,d\}$.

Solution : Let $A = \{a,b,c,d\}$

P(A) = { ϕ , {*a*}, {b} , {c}, {d} , {*a*,b}, {*a*,c} , {*a*,d} , {b,c} , {b,d} , {c,d} {*a*,b,c}, {*a*,b,d}, {*a*,c,d}, {b,c,d} , {*a*,b,c,d}}



4. Prove that for any three sets A,B,and C,

(i)A× (B∪C) = (A×B) ∪ (A× C)

(ii)A× (B∩C) = (A×B) ∩ (A× C)

Solution :

(i) $A \times (B \cup C) = \{ (x,y) \mid (x \in A) \text{ and } (y \in B \cup C)\}$

$= \{ (x,y) \mid (x \in A) \text{ and } (y \in B) \text{ or } (y \in C)\}$

$= \{ (x,y) \mid ((x \in A) \text{ and } (y \in B)) \text{ or } ((x \in A) \text{ and } (y \in C))\}$

$= (A \times B) \cup (A \times C)$

(ii) $A \times (B \cap C) = \{ (x,y) \mid (x \in A) \text{ and } (y \in B \cap C)\}$

$= \{ (x,y) \mid (x \in A) \text{ and } (y \in B) \text{ and } (y \in C)\}$

$=\{ (x,y) \mid ((x \in A) \text{ and } (y \in B)) \text{ and } ((x \in A) \text{ and } (y \in C))\}$

$= (A \times B) \cap (A \times C)$

5. Let X = { 1,2,...,7} and R = { (x,y) | x-y is divisible by 3}. Show that R is an equivalence relation. Draw the graph of R.

Solution : Let X = {1,2,3,4,5,6,7}

R = { (x,y) | x-y is divisible by 3}

x-x = 0 is divisible by 3.

Therefore, $(x,x) \in R$ for all $x \in X$

Therefore, R is reflexive

Let $(x,y) \in R$

x-y is divisible by 3

-(x-y) is divisible by 3

y-x is divisible by 3

Therefore , $(y,x) \in R$

Therefore, R is symmetric

Let $(x,y) \in R$ and $(y,z) \in R$

$(x,y) \in R$ implies x-y is divisible by 3

$(y,z) \in R$ imples y-z is divisible by 3

(x-y) + (y-z) = (x-z) is divisible by 3

Therefore, $(x,z) \in R$

Therefore, R is transitive.

Therefore, R is an equivalence relation.

6. Given S = {1,2,…,10} and a relation R on S where R = { (x,y) | x +y =10}. What are the properties of the relation R ?

Solution :

Let S = {1,2,3,4,5,6,7,8,9,10}

R = { (x,y) | x +y =10}

R = { (1,9), (2,8), (3,7), (4,6),(5,5), (6,4), (7,3), (8,2),(9,1)}

R is not reflexive (because $(1,1) \notin R$)

R is symmetric

R is not transitive (since, $(1,9) \in R$, $(9,1) \in R$ but $(1,1) \notin R$)

R is not irreflexive (since, $(5,5) \in R$)

R is not antisymmetric (since, $(1,9) \in R$, $(9,1) \in R$ but 1 is not equal to 9)

R is not an equivalence relation

7. If R is a partial ordering relation on a set X and $A \subseteq X$, show that $R \cap (A \times A)$ is a partial ordering relation on A .

Solution :

Let R be a partial ordering relation on a set X and $A \subseteq X$

R is a partial ordering relation $\Rightarrow$ R is reflexive, antisymmetric and transitive

R is reflexive $\Rightarrow$ $(x,x) \in R$ for all $x \in X$

$\Rightarrow$ $(x,x) \in R$ for all $x \in A$ (since, $A \subseteq X$)

$x \in A \Rightarrow (x,x) \in A \times A$ for all $x \in A$

Therefore, $(x,x) \in R \cap (A \times A)$ for all $x \in A$

Therefore, $R \cap (A \times A)$ is reflexive on A

Let $(x,y) \in R \cap (A \times A)$ and $(y,x) \in R \cap (A \times A)$

$(x,y) \in R$, $(y,x) \in R$ and R is antisymmetric $\Rightarrow x = y$

Therefore, $R \cap (A \times A)$ is antisymmetric on A

Let $(x,y) \in R \cap (A \times A)$ and $(y,z) \in R \cap (A \times A)$

$(x,y) \in R$, $(y,z) \in R$ and R is transitive $\Rightarrow (x,z) \in R$ for all $x,y,z \in A$

$(x,y) \in A \times A$,and $(y,z) \in A \times A \Rightarrow (x,z) \in A \times A$ for all $x,y,z \in A$

Therefore, $(x,z) \in R \cap (A \times A)$

Therefore, $R \cap (A \times A)$ is transitive on A

Therefore, $R \cap (A \times A)$ is reflexive, antisymmetric, and transitive on A

Therefore, $R \cap (A \times A)$ is a partial ordering relation on A .

8. Let $X = \{2,3,6,12,24,36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y. Draw the Hasse diagram of $(X, \leq)$.

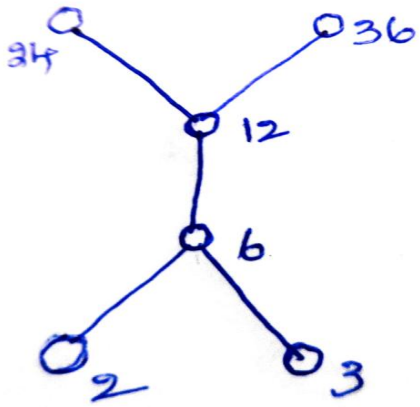
Solution :

Let $X = \{2,3,6,12,24,36\}$

Given that $x \leq y$ if x divides y

Divides is a partial ordering relation on X. and $(X, \leq)$ is a poset

The Hasse diagram of $(X, \leq)$ is



9. Show that the relation R = {(a, b) | a ≡ b (mod m)}, where m be an integer with m > 1 is an equivalence relation on the set of integers.

Solution : Let R = {(a, b) | a ≡ b (mod m)}, where m be an integer with m > 1 be a relation on the set of integers.

a ≡ b (mod m) ⇒ m / (a-b) ⇒ There exists an integer q such that (a-b) = mq.

We have , a- a = 0 for any integer a and m / 0

Therefore, a ≡ a (mod m) ⇒ (a,a) ∈ R for every integer

⇒ R is reflexive on set of integers

Let (a,b) ∈ R for integers a,b.

⇒ a ≡ b (mod m) ⇒ m / (a-b) ⇒ m / -(a-b) ⇒ m / (b-a) ⇒ b ≡ a (mod m)

⇒(b,a) ∈ R

Therefore, R is symmetric.

Let (a,b) ∈ R and (b,c) ∈ R for integers a,b,c

(a,b) ∈ R ⇒ a ≡ b (mod m) ⇒ m / (a-b) ⇒ There exists an integer q_1 such that (a-b) = mq_1.

(a-b) = m q_1 → (1)

(b,c) ∈ R ⇒ b ≡ c (mod m) ⇒ m / (b-c) ⇒ There exists an integer q_2 such that (b-c) = mq_2.

(b-c) = mq_2 → (2)

Eq.(1) + Eq.(2) ⇒ (a-b) + (b-c) = a-c = m q_1 + mq_2 = m (q_1 + q_2) = m q_3 where q_3= (q_1 + q_2) is an integer.

Therefore, a-c = m q_3 ⇒ m/ (a-c) ⇒ a ≡ c (mod m)

⇒(a,c) ∈ R

Therefore, (a,b) ∈ R and (b,c) ∈ R ⇒(a,c) ∈ R

Therefore , R is transitive.

Therefore, R is reflexive, symmetric, and transitive.

Therefore , R is an equivalence relation.

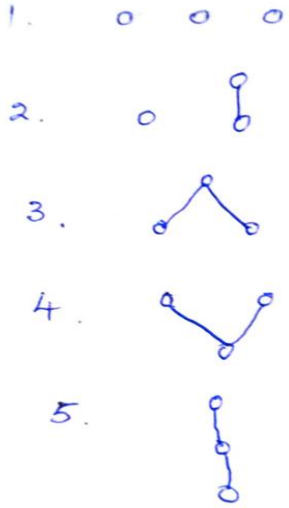
10. Let R denote a relation on the set of ordered pairs of positive integers such that (x,y) R (u,v) iff xv=yu. Show that R is an equivalence relation.

Solution :

Let R denote a relation on the set of ordered pairs of positive integers such that (x,y) R (u,v) iff xv=yu

xx = xx implies (x,x) R (x,x) for all ordered pairs of positive integers.
Therefore, R is reflexive.
Let (x,y) R (u,v)
Therefore xv = yu implies vx = uy implies uy = vx implies (u,v) R (x,y)
Therefore, R is symmetric
Let (x,y) R (u,v) and (u,v) R (p,s)
(x,y) R (u,v) implies xv = yu
(u,v) R (p,s) implies us = vp
To prove (x,y) R (p,s) it is enough to prove xs = yp
xv = yu →1
x = yu/v
us = vp →2
u = vp/s
Multiplying Eq. (1) by s
xvs = yus
xvs = y (vp/s) s (From (2))
xvs = yvp
x(sv) =y (pv) (Associative and commutative)
(xs) v = (yp) v (associative)
Therefore, xs = yp (by cancellation)
Therefore, (x,y) R (p,s)
Therefore, R is transitive.
Therefore, R is reflexive, symmetric , and transitive
Therefore, R is an equivalence relation.

11. Show that there are only five distinct Hasse diagrams for partially ordered sets that contain three elements.
Solution :



12. Define the following in a Poset
    (i) least member
    (ii) greatest member
    (iii) minimal member
    (iv) maximal member

Solution :

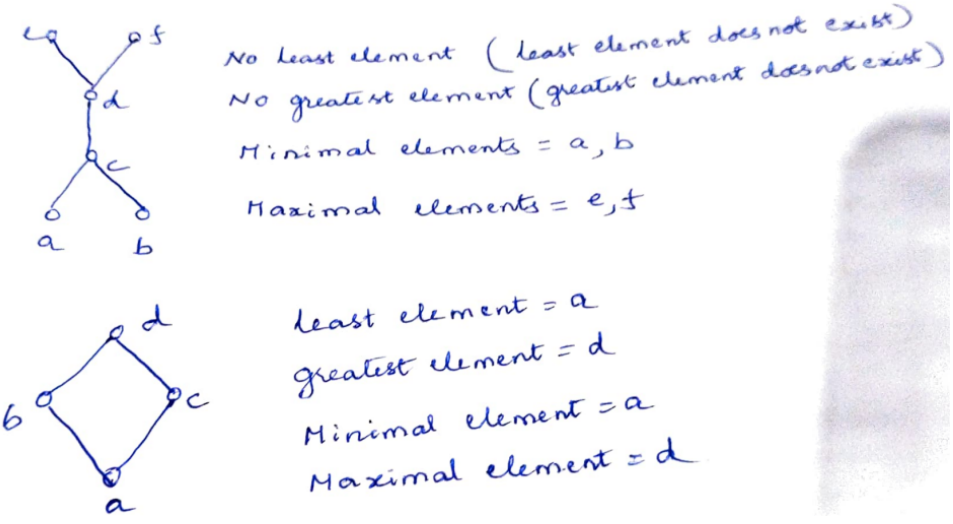
A partially ordered set (Poset) is a set with a partial order relation, often denoted as $\leq$ or $\preceq$.

(i) Least Member: A least member (or minimum element) of a Poset is an element that is less than or equal to every other element in the set. If it exists, it is unique.

(ii) Greatest Member: A greatest member (or maximum element) of a Poset is an element that is greater than or equal to every other element in the set. If it exists, it is unique.

(iii) Minimal Member: A minimal member (or minimal element) of a Poset is an element that is not greater than any other element in the set. In other words, there is no other element less than it. A Poset can have multiple minimal elements.

(iv) Maximal Member: A maximal member (or maximal element) of a Poset is an element that is not less than any other element in the set. In other words, there is no other element greater than it. A Poset can have multiple maximal elements.



13. Discuss the concept of functions, including definitions, domain, range, and types of functions.

Solution :

Function: A function is a relation between a set of inputs, called the domain, and a set of possible outputs, called the range. It assigns each input exactly one output.

Domain: The set of all possible input values for a function.
Range: The set of all possible output values for a function.
Codomain: The set of all possible output values that a function can take.

Types of Functions:

1. **One-to-One (Injective) Function**: Each output value corresponds to exactly one input value.
2. **Onto (Surjective) Function**: Every element in the codomain is an output of the function for some input.
3. **Bijective Function:** A function that is both one-to-one and onto.
4. **Constant Function:** A function that assigns the same output value to every input value.
5. **Identity Function:** A function that assigns each input value to itself.

Key Concepts :

Function Notation: A way of writing functions using the notation f(x), where f is the name of the function and x is the input value.

Function Composition: The process of combining two or more functions to create a new function.

Importance of Functions : Functions are used to model real-world relationships and phenomena. They are essential in mathematics, science, and engineering. Functions help to analyze and solve problems in various fields.

14. Let X = {1, 2, 3} and f, g, h and s are functions from X to X defined by
f = {(1, 2), (2, 3), (3, 1)}, g = {(1, 2), (2, 1), (3, 3)}, h = {(1, 1), (2, 2), (3, 1)} s = {(1, 1), (2, 2), (3, 3)}. Then find fog, fos, fohog and sog.

Solution :

Given that X = {1, 2, 3}, f = {(1, 2), (2, 3), (3, 1)}, g = {(1, 2), (2, 1), (3, 3)},
h = {(1, 1), (2, 2), (3, 1)}, s = {(1, 1), (2, 2), (3, 3)}

(fog)(x) = f(g(x))

(fog)(1) = f(g(1)) = f(2) = 3
(fog)(2) = f(g(2)) = f(1) = 2
(fog)(3) = f(g(3)) = f(3) = 1

Therefore, fog = {(1, 3), (2, 2), (3, 1)}

(fos)(x) = f(s(x))

(fos)(1) = f(s(1)) = f(1) = 2
(fos)(2) = f(s(2)) = f(2) = 3
(fos)(3) = f(s(3)) = f(3) = 1

Therefore, fos = {(1, 2), (2, 3), (3, 1)}

(fohog)(x) = f(h(g(x)))

(fohog)(1) = f(h(g(1))) = f(h(2)) = f(2) = 3
(fohog)(2) = f(h(g(2))) = f(h(1)) = f(1) = 2
(fohog)(3) = f(h(g(3))) = f(h(3)) = f(1) = 2

Therefore, fohog = {(1, 3), (2, 2), (3, 2)}

(sog)(x) = s(g(x))

(sog)(1) = s(g(1)) = s(2) = 2
(sog)(2) = s(g(2)) = s(1) = 1
(sog)(3) = s(g(3)) = s(3) = 3

Therefore, sog = {(1, 2), (2, 1), (3, 3)}

15. Explain the concept of inverse functions in detail, including the conditions for a function to be invertible.
Solution :

Inverse Functions: An inverse function is a function that "reverses" another function. Given a function f(x), its inverse is denoted as $f^{-1}(x)$, and it satisfies the following property:

$f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$

Conditions for Invertibility: A function f(x) is invertible if and only if it is:

1. One-to-One (Injective): Each output value corresponds to exactly one input value.
2. Onto (Surjective): Every element in the codomain is an output of the function for some input.

Finding the Inverse of a Function: To find the inverse of a function f(x), follow these steps:

1. Write the function as $y = f(x)$.
2. Swap x and y: Write $x = f(y)$.
3. Solve for y: Express y in terms of x.
4. Write the inverse function as $f^{-1}(x) = y$.

Example: Consider the function $f(x) = 2x + 3$. To find its inverse:

1. Write $y = 2x + 3$.
2. Swap x and y: $x = 2y + 3$.
3. Solve for y: $y = (x - 3)/2$.
4. Write the inverse function: $f^{-1}(x) = (x - 3)/2$.

Properties of Inverse Functions:
1. **Uniqueness:** If a function is invertible, its inverse is unique.
2. **Symmetry:** The graph of $f^{-1}(x)$ is symmetric to the graph of $f(x)$ with respect to the line $y = x$.
3. **Composition:** $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

16. Let $f : R \to R$ and $g : R \to R$, where R is the set of real numbers. Then find fog and gof where $f(x) = x^2 - 2$ and $g(x) = x + 4$. State whether these functions are injective, surjective, and bijective.
Solution :

Let $f : R \to R$ and $g : R \to R$, where R is the set of real numbers.

$f(x) = x^2 - 2$
$g(x) = x + 4$

$(fog)(x) = f(g(x))$
$= f(x + 4)$
$= (x + 4)^2 - 2$
$= x^2 + 8x + 16 - 2$
$= x^2 + 8x + 14$

$(gof)(x) = g(f(x))$
$= g(x^2 - 2)$
$= (x^2 - 2) + 4$
$= x^2 + 2$

$f(x) = x^2 - 2$
Injective : f(x) is not injective because $f(-x) = f(x)$ for all x.
Surjective : f(x) is not surjective because its range is $[-2, \infty)$, which is not equal to R.
Bijective: f(x) is not bijective because it is neither injective nor surjective.

$g(x) = x + 4$
Injective : g(x) is injective because g(a) = g(b) implies a = b.
Surjective : g(x) is surjective because its range is R. Co-domain = Range
Bijective : g(x) is bijective because it is both injective and surjective.

$fog(x) = x^2 + 8x + 14$
Injectivity: fog(x) is not injective because it is a quadratic function.
Surjective : fog(x) is not surjective because its range is not R.
Bijectivity: fog(x) is not bijective because it is neither injective nor surjective.

$gof(x) = x^2 + 2$
Injective : gof(x) is not injective because gof(-x) = gof(x) for all x.
Surjective : gof(x) is not surjective because its range is $[2, \infty)$, which is not equal to R.
Bijective : gof(x) is not bijective because it is neither injective nor surjective.

17. Define one-one and onto functions. Show that the functions $f(x) = x^2$ and $g(x) = x^{1/2}$, for $x \in [0, \infty)$, are inverse of one another.

Solution :
One-One (Injective) Function: A function f: A → B is said to be one-one or injective if for every element b in B, there is at most one element a in A such that f(a) = b. In other words, distinct elements in A map to distinct elements in B.

Onto (Surjective) Function: A function f: A → B is said to be onto or surjective if for every element b in B, there exists an element a in A such that f(a) = b. In other words, every element in B is mapped to by at least one element in A.

Inverse Functions: Two functions f and g are inverses of each other if:

(i) (fog)(x) = x for all x in the domain of g
(ii) (gof)(x) = x for all x in the domain of f

$(fog)(x) = f(g(x)) = f(x^{(1/2)}) = (x^{(1/2)})^2 = x$
$(gof)(x) = g(f(x)) = g(x^2) = (x^2)^{(1/2)} = x$ (since $x \geq 0$)

Since (fog)(x) = x and (gof)(x) = x for x in $[0, \infty)$, we can say that $f(x) = x^2$ and $g(x) = x^{(1/2)}$ are inverses of each other over the domain $[0, \infty)$.